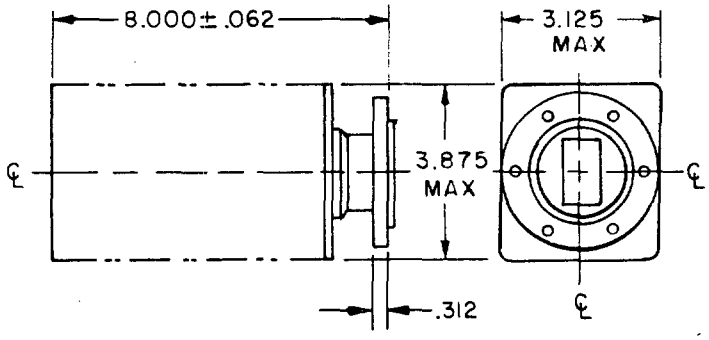
MIL-D-3954/16A
29 May 1974
SUPERSEDING
MIL-D-3954/16
14 October 1966, and
MS35891
15 May 1962

MILITARY SPECIFICATION SHEET

# DUMMY LOADS, ELECTRICAL, WAVEGUIDE
(FREQUENCY RANGE 5.85 TO 8.20 GIGAHERTZ)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dummy loads described herein shall consist of this document and the latest issue of Specification MIL-D-3954.




| INCHES | MM |
|---|---|
| .062 | 1.57 |
| .312 | 7.92 |
| 3.125 | 79.38 |
| 3.875 | 98.43 |
| 8.00 | 203.20 |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.
5. Quantity of fins required will be dependent on heat level requirement of the load.

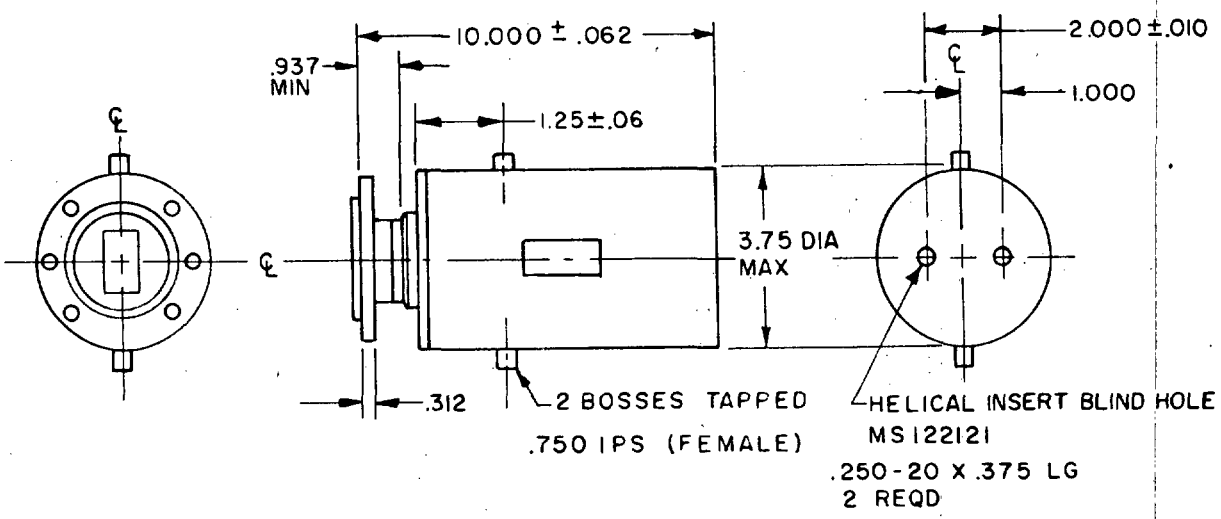
Ⓐ FIGURE 1. Class I dummy load.

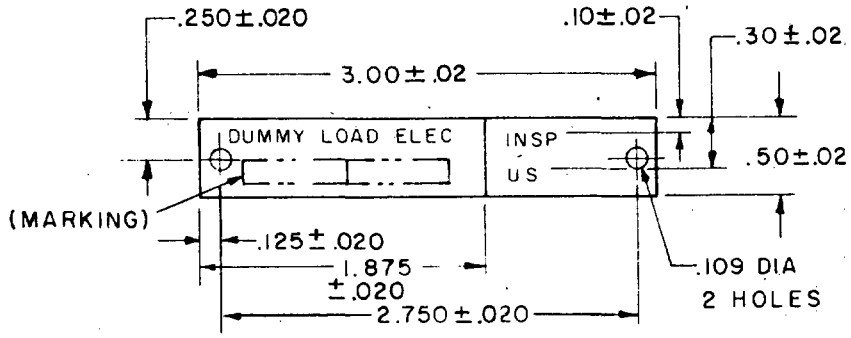
Ⓐ denotes changes.



FSC 5985

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IDENTIFICATION PLATE

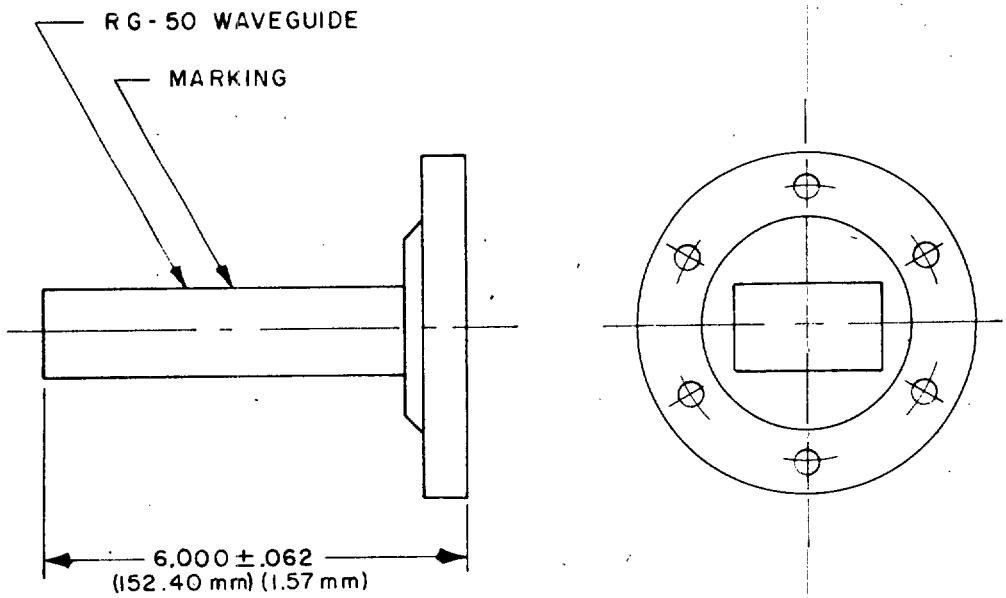
| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|
| .010 | .25 | .125 | 3.18 | .750 | 19.05 | 2.750 | 69.85 |
| .02 | .51 | .250 | 6.35 | .937 | 23.80 | 3.00 | 76.20 |
| .06 | 1.52 | .30 | 7.62 | 1.000 | 25.40 | 3.75 | 95.25 |
| .062 | 1.57 | .312 | 7.92 | 1.25 | 31.75 | 10.000 | 254.00 |
| .10 | 2.54 | .375 | 9.53 | 1.875 | 47.63 | | |
| .109 | 2.77 | .50 | 12.70 | 2.000 | 50.80 | | |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.

Ⓐ FIGURE 2. Class II dummy load.



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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Screws, lockwashers and gasket are supplied with mating flange.

Ⓐ FIGURE 3. Class III dummy load.



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TABLE I. Characteristics.

| Part number | Class Ⓐ | Material | Flange equal to Ⓐ | VSWR max | Power | | Pressure | | Figure | Flow rate | Input temp. | Output temp. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Average | Peak | Internal | Coolant chamber | | | | |
| | | | | | (watts) | (kilowatts) | (psig) | (psig) | | (GPM 1/) | (°C) | (°C) |
| M3954/16-01 | I | AL | 2/ M3922/55-002 (UG-441/U) | 1.10 | 800 3/ | 710 | 30 | | 1 | | | |
| M3954/16-02 | II | AL | 2/ M3922/55-002 (UG-441/U) | 1.10 | 2000 | 710 | 30 | 100 | 2 | 0.667 | 35 | 60 |
| M3954/16-03 | II | Copper | 2/ M3922/55-001 (UG-344/U) | 1.10 | 2000 | 710 | 30 | 100 | 2 | 0.667 | 35 | 60 |
| M3954/16-04 | II | Corrosion resisting steel | 2/ M3922/55-001 (UG-344/U) | 1.10 | 2000 | 710 | 30 | 100 | 2 | 0.667 | 35 | 60 |
| Ⓐ M3954/16-05 | III | Copper | M3922/55-001 (UG-344/U) | 1.10 | 2 | 2 | | | 3 | | | |

1/ The flow rate was determined from the following formula:

$$Q = \frac{6.8P}{Cp\ \Delta T}$$

Where: Q = Minimum flow rate in GPM.
P = Avg power in kilowatts.
Cp = Specific heat of coolant.
$\Delta T$ = Coolant temperature rise in °F.

The calculations were made for Cp = 1 for water, a $\Delta T$ of 45°F and a safety factor of approximately 2. For different coolants or different temperature rises, a different flow rate would be necessary.

2/ Except thickness.

3/ This is a test power value. The average rated power is 1,000 watts.

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(A) TABLE II. Cross reference part number/ AN nomenclature.

| Part number | AN nomenclature |
|---|---|
| M3954/16-01 | DA-144/U |

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figures 1, 2, and 3.
Weight:
Dry load - 4.1 lbs max.
Liquid cooled load - 5 lbs max.

Performance characteristics: See table I.

(A) Shock (for part no. M3954/16-05 only): Method 213 of MIL-STD-202, condition A.

Part number: M3954/16- (dash number from table I).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - CG, MC, AS, OS, SH
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0754-6)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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